Diagonal Gaussian forcing: trace thresholds, Wick centering and recovery geometry

Project Stardust

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Abstract

Countably many independent exponentially damped Gaussian modes define an actual square-integrable Hilbert-valued field at positive time exactly when the squared forcing amplitudes divided by the damping rates are summable. We prove this criterion, the exact covariance and cutoff error, and mean-square continuity up to time zero. The field is a closed linear Gaussian field in the original source space; its full Gaussian divergence exists, lies in second chaos, and has norm at most $\sqrt{2}$ times the field norm. The proof reconstructs finite Gaussian integration by parts and Hilbert–Schmidt completion, including the symmetric/antisymmetric kernel distinction. In contrast, observation of all Brownian modes at any positive time retains an infinite first-chaos isometry, preventing compact divergence recovery. We also prove that the linear-rate deterministic synthesis is compact but has infinite stochastic trace. Wick-centered observables can converge when the corresponding raw sums fail; their exact Brownian prediction and increment energies distinguish initial Hermite scaling from positive-time roughness. Predictions from arbitrary Gaussian observations retain full projected Gram cross terms. Normalized Wick metric bounds retain the positive-correlation hypothesis; fractional positive-time exponents and the antipodal obstruction are proved explicitly. These conclusions concern distinct operators. The trace and Gaussian mechanisms are classical; the paper consolidates frozen verified constructions and explicit counterexamples without claiming new stochastic PDE theory.

1 The actual modes and the trace criterion

Let $\mathcal{H}$ be a real separable infinite-dimensional Hilbert space with an isonormal Gaussian map $X : \mathcal{H} \to L^2(P)$: $\mathbb{E}X(h) = 0$, $\mathbb{E}[X(h)X(k)] = \langle h, k\rangle$, and finite families are jointly Gaussian. All scalar and field L^2 spaces below are over the Gaussian sigma-algebra generated by $X(\mathcal{H})$, modulo null sets; no additional non-Gaussian random factor is included. Embed mutually orthogonal copies of $L^2(\mathbb{R}_+)$ isometrically into $\mathcal{H}$, with embeddings ι_n, $n \geq 0$. Thus $B_n(t) = X(\iota_n \mathbf{1}_{(0,t]})$ are independent Brownian coordinates at the level of their joint Gaussian laws. Continuous versions give the usual Brownian realization if desired; only the stated L^2 coordinates are used below.

Fix deterministic rates $\lambda_n \geq 1$, amplitudes $c_n \in \mathbb{R}$, and an orthonormal sequence (e_n) in the same source coefficient space $\mathcal{H}$. Define

$$k_{n,t}(s) = \mathbf{1}_{(0,t]}(s) e^{-\lambda_n (t-s)}, \quad h_{n,t} = \iota_n k_{n,t}, \quad Y_n(t) = X(h_{n,t}). \tag{1}$$

The deterministic-integrand Brownian notation $Y_n(t) = \int_0^t e^{-\lambda_n(t-s)}\, dB_n(s)$ is equivalent to this isonormal definition. No adaptedness of the coefficient vectors e_n relative to the noise is needed. They are deterministic output directions; they need not be orthogonal to all input directions.

For $N \geq 1$ put

$$F_N(t) = \sum_{n<N} c_n e_n Y_n(t) \in L^2(P; \mathcal{H}).$$

This is a Hilbert-valued random field, not a scalar sum and not yet a claimed solution of an arbitrary SPDE.

Theorem 1.1. *For each $t > 0$, the following are equivalent:*

1. *$F_N(t)$ converges in $L^2(P; \mathcal{H})$;*
2. $\displaystyle\sum_{n\geq 0} \frac{c_n^2}{\lambda_n} < \infty.$

When these hold, write the limit as $F(t)$. Then

$$\|F(t)\|_2^2 = \sum_{n\geq 0} c_n^2 \frac{1 - e^{-2\lambda_n t}}{2\lambda_n}, \tag{2}$$

$$\|F(t) - F_N(t)\|_2^2 = \sum_{n\geq N} c_n^2 \frac{1 - e^{-2\lambda_n t}}{2\lambda_n} \leq \frac{1}{2} \sum_{n\geq N} \frac{c_n^2}{\lambda_n}. \tag{3}$$

Setting $F(0) = 0$, the curve $t \mapsto F(t)$ is continuous in $L^2(P; \mathcal{H})$ on $[0, \infty)$.

The reciprocal-rate condition is a diagonal instance of the classical stochastic-convolution Hilbert–Schmidt criterion; see Hairer's primary lecture notes [3], Sections 4.4 and 6. The theorem here fixes the actual source construction and distinguishes its positive-time quantifier from the trivial zero-time field.

2 Finite-mode identities and completion

Lemma 2.1. *For $\lambda > 0$, let $k_{\lambda,t}(r) = \mathbf{1}_{(0,t]}(r) e^{-\lambda(t-r)}$ and $V_\lambda(t) = (1 - e^{-2\lambda t})/(2\lambda)$. Then*

$$\|k_{\lambda,t}\|_{L^2}^2 = V_\lambda(t), \tag{4}$$

$$\langle k_{\lambda,t}, k_{\lambda,s} \rangle = e^{-\lambda|t-s|} V_\lambda(\min(s,t)), \tag{5}$$

$$\|k_{\lambda,t} - k_{\lambda,s}\|_{L^2}^2 = V_\lambda(t) + V_\lambda(s) - 2e^{-\lambda|t-s|} V_\lambda(\min(s,t)). \tag{6}$$

For $\lambda \geq 1$,

$$\frac{1 - e^{-2t}}{2\lambda} \leq V_\lambda(t) \leq \frac{1}{2\lambda}, \qquad \|k_{\lambda,t} - k_{\lambda,s}\|^2 \leq \frac{2}{\lambda}. \tag{7}$$

Proof. Integrating $e^{-2\lambda(t-r)}$ over $0 < r \leq t$ proves (4). If $s \leq t$, factor $e^{-\lambda(t-s)}$ from the overlap integral on $(0, s]$ and use (4); symmetry gives (5). Expansion of a squared norm proves (6). The first bounds follow from $0 \leq e^{-2\lambda t} \leq e^{-2t} \leq 1$. Finally

$$\|k_{\lambda,t} - k_{\lambda,s}\|^2 \leq 2\|k_{\lambda,t}\|^2 + 2\|k_{\lambda,s}\|^2 \leq 2/\lambda.$$

□

Proof of Theorem 1.1. Orthonormality of the deterministic output vectors gives, for any finite index set A,

$$\left\| \sum_{n\in A} c_n e_n Y_n(t) \right\|_2^2 = \sum_{n\in A} c_n^2 \mathbb{E} Y_n(t)^2 = \sum_{n\in A} c_n^2 V_{\lambda_n}(t). \tag{8}$$

In particular, disjoint finite index sets give orthogonal fields. Thus the partial sums are Cauchy exactly when the nonnegative energy series converges. At fixed $t > 0$, the factor $1 - e^{-2t}$ is positive, so (7) makes that convergence equivalent to the reciprocal-rate series. Completeness gives $F(t)$. Passing finite orthogonal sums to their limits gives (2) and (3), including the stated upper error bound.

For continuity fix $s \geq 0$. Each right side in (6) is continuous in t and vanishes at $t = s$. Consequently $k_{\lambda_n,t} \to k_{\lambda_n,s}$ in L^2 for each n. Orthogonality gives

$$\|F(t) - F(s)\|_2^2 = \sum_{n\geq 0} c_n^2 \|k_{\lambda_n,t} - k_{\lambda_n,s}\|_{L^2}^2. \tag{9}$$

For example, this equality follows first for F_N, then by its convergence at t and s and the dominated summability in (7). The summable bound $2c_n^2/\lambda_n$ permits dominated convergence of the series, so the squared increment norm tends to zero. This also proves continuity at $s = 0$. □

When $c_n = 1$, the rates $(n+1)^2$ admit the field at all times, while $\lambda_n = n + 1$ do not admit the L^2 series at any positive time. Indeed the reciprocal square series converges, for example by comparison with the telescoping series $1/(n(n+1))$ for $n \geq 1$; the harmonic series diverges by grouping indices in dyadic blocks. For an undamped unit Brownian series the cutoff energy is Nt. Zero damping lies outside the hypothesis $\lambda_n \geq 1$ and is this separate elementary example, not a substitution into (2) with a zero denominator.

These statements assert L^2 convergence and mean-square continuity. They do not assert a samplewise uniform-in-time limit, a path-space Hölder exponent, or a continuum nonlinear PDE solution.

3 The original Gaussian divergence on linear fields

Let D be the original closed Gaussian gradient, obtained by closing the polynomial cylindrical gradient in $L^2(P) \to L^2(P; \mathcal{H})$. Its maximal Hilbert adjoint is the Gaussian divergence $\delta = D^*$. The relevant conventions agree with [4]. Write $\mathcal{L}$ for the closed linear span in $L^2(P; \mathcal{H})$ of fields $aX(b)$ with $a, b \in \mathcal{H}$. This is the space of linear Gaussian fields, not the full random field space.

Lemma 3.1. *For $a, b \in \mathcal{H}$,*

$$aX(b) \in D(\delta), \qquad \delta(aX(b)) = X(a)X(b) - \langle a, b\rangle. \tag{10}$$

For a finite-rank deterministic operator $K = \sum_{j=1}^m a_j \otimes b_j$, where $(a \otimes b)h = a\langle b, h\rangle$, put $F_K = \sum_j a_j X(b_j)$. Then

$$\|F_K\|_2^2 = \|K\|_{\mathrm{HS}}^2, \tag{11}$$

$$\|\delta F_K\|_2^2 = \|K\|_{\mathrm{HS}}^2 + \langle K, K^*\rangle_{\mathrm{HS}} = 2\|\operatorname{sym} K\|_{\mathrm{HS}}^2 \leq 2\|F_K\|_2^2. \tag{12}$$

Here $\operatorname{sym} K = (K + K^*)/2$. *The divergence is in second Gaussian chaos.*

Proof. For a polynomial cylindrical test f, Gaussian integration by parts applied to the product $fX(b)$ gives

$$\mathbb{E}[X(a)fX(b)] = \mathbb{E}[\langle Df, a\rangle X(b)] + \langle a, b\rangle \mathbb{E}f.$$

This proves (10) on the core. Passing graph approximants of f through the pairing extends it to every $f \in D(D)$; the right side of (10) is in L^2 by Gaussian moments. It therefore meets the defining maximal-adjoint bound.

The isonormal covariance gives $\mathbb{E}\langle F_K, F_L\rangle = \langle K, L\rangle_{\mathrm{HS}}$, proving (11) and representation independence. For the divergence, the centered Gaussian fourth-moment identity gives

$$\begin{aligned}\mathbb{E}[(X(a)X(b) - \langle a, b\rangle)(X(c)X(d) - \langle c, d\rangle)] \\ = \langle a, c\rangle\langle b, d\rangle + \langle a, d\rangle\langle b, c\rangle.\end{aligned}$$

That identity follows by differentiating the finite joint Gaussian exponential moment formula four times at zero. Summing proves the first equality of (12). Orthogonality of the symmetric and skew-adjoint parts in Hilbert–Schmidt inner product proves the second equality and the upper bound. The centered quadratic products are degree-two Hermite polynomials or their polarizations, hence lie in second chaos. □

Theorem 3.2. *Every field $F \in \mathcal{L}$ belongs to the original domain $D(\delta)$. The restricted divergence is bounded, with*

$$\|\delta F\|_2 \le \sqrt{2}\|F\|_2. \tag{13}$$

It maps into second chaos. For $F = F_K$ with K Hilbert–Schmidt, its kernel consists exactly of skew-adjoint K. Its canonical gradient recovery is $F_{\operatorname{sym} K}$ and

$$R(\delta F_K) = F_{\operatorname{sym} K}, \qquad \|\delta F_K\|_2^2 = 2\|F_{\operatorname{sym} K}\|_2^2. \tag{14}$$

Proof. The finite-rank isometry (11) extends by Hilbert–Schmidt completion onto $\mathcal{L}$. For finite-rank approximants $K_j \to K$, Lemma 3.1 shows that δF_{K_j} is Cauchy, with the bound (13). Closedness of the original adjoint therefore puts $(F_K, \lim_j \delta F_{K_j})$ in its original graph. Second chaos is closed, so the limit remains in it. The exact norm formula (12) passes to the limit, proving the kernel and energy assertions.

On centered Gaussian chaos define the canonical recovery as $R = DN^{-1}$, where the number operator N has eigenvalue q on chaos $q \ge 1$, and set $R1 = 0$. On second chaos it is $D/2$. Finite differentiation gives

$$D\delta F_K = F_{K+K^*} = 2F_{\operatorname{sym} K}$$

for finite-rank K. Passing to the original closed gradient graph proves the identity for Hilbert–Schmidt K, since both sides converge strongly. This gives (14). □

For clarity, the recovery used here is bounded on full scalar L^2: if $f = \sum_{q\ge 0} f_q$ is its orthogonal chaos decomposition, the finite Gaussian identity $\|Df_q\|_2^2 = q\|f_q\|_2^2$ gives

$$\|Rf\|_2^2 = \sum_{q\ge 1} \frac{\|f_q\|_2^2}{q} \le \|f\|_2^2.$$

The finite identity follows from the one-coordinate Hermite derivative and orthogonality, then tensorization. Different differentiated chaos orders are orthogonal. Thus the series defines R by norm completion. In the frozen library it is constructed as the minimal-norm original divergence

recovery; its chaos identity identifies it with this DN^{-1}. Only its proved action in (14) and on first chaos is needed in this paper.

The constant $\sqrt{2}$ is attained for a unit $a = b$ in (10). Antisymmetric finite tensors instead give zero divergence with a nonzero field. Thus no isometry on all of $\mathcal{L}$ is asserted, and scalar divergence does not recover the antisymmetric component of a field.

4 Divergence and limits of the admitted forcing

Under the trace criterion of Theorem 1.1, define

$$K_t = \sum_{n\geq 0} c_n e_n \otimes h_{n,t}.$$

Orthonormality of e_n makes this an actual Hilbert–Schmidt sum: $\|K_t\|_{\mathrm{HS}}^2 = \sum_n c_n^2 \|h_{n,t}\|^2$. It follows that $F(t) = F_{K_t} \in \mathcal{L}$.

Corollary 4.1. *For every admitted forcing profile, $F(t) \in D(\delta)$ at every $t \geq 0$, and*

$$\delta F_N(t) \longrightarrow \delta F(t) \quad \text{in } L^2(P), \qquad \|\delta F(t) - \delta F_N(t)\|_2 \leq \sqrt{2}\|F(t) - F_N(t)\|_2.$$

The curve $t \mapsto \delta F(t)$ is mean-square continuous and has the exact symmetric recovery (14) *at each time.*

Proof. Apply Theorem 3.2 to $F(t)$ and to the differences of finite or limiting fields in $\mathcal{L}$. The norm estimate proves the convergence. Applying the same estimate to $F(t) - F(s)$ and using Theorem 1.1 proves continuity. Recovery follows from the identity for K_t. □

This divergence is the original Gaussian source adjoint, not an Itô integral in the physical time of F. In particular the field's deterministic output coefficient vectors may overlap its source directions, and the centered quadratic correction in (10) must be retained. The trace criterion controls the specific field; it does not make every observation or recovery map compact.

5 Infinitely many observed modes and noncompact recovery

Let the complete modal past at time t be

$$\mathcal{A}_t = \sigma(B_n(s) : n \geq 0,\ 0 \leq s \leq t), \qquad Q_t = \mathbb{E}[\,\cdot \mid \mathcal{A}_t].$$

These are the actual Gaussian observation spaces of all modes, independent of whether a particular amplitude/rate profile is trace-admitted.

Theorem 5.1. *The bounded operator $RQ_t : L^2(P) \to L^2(P;\mathcal{H})$ is compact if and only if $t = 0$. At every positive t, even recovery restricted to the observed first chaos contains an infinite-dimensional isometry.*

Proof. For $t > 0$ set $u_n = t^{-1/2}\iota_n \mathbf{1}_{(0,t]}$. These form an orthonormal sequence in $\mathcal{H}$, and $X(u_n)$ form an orthonormal sequence in $L^2(P)$. Each is observed at time t, so $Q_t X(u_n) = X(u_n)$. On first chaos the number operator is one and $DX(u_n) = u_n$, a deterministic coefficient vector. Therefore

$$RQ_t X(u_n) = u_n \quad \text{as a constant field in } L^2(P;\mathcal{H}).$$

These outputs are pairwise at distance $\sqrt{2}$. A compact operator would take this bounded input sequence to a sequence with a norm convergent subsequence, a contradiction. Taking closed linear spans also proves the asserted isometry. At $t = 0$ all modal observations are zero; Q_0 projects onto constants, which R kills. Thus $RQ_0 = 0$ is compact. □

The obstruction does not depend on the chosen canonical right inverse. Indeed, if a bounded linear map L satisfies $\delta L(X(u)) = X(u)$ for every u in an infinite observed coefficient subspace, adjointness against $X(v)$ gives

$$\langle \mathbb{E}L(X(u)), v\rangle = \mathbb{E}\langle L(X(u)), DX(v)\rangle = \langle u, v\rangle.$$

Thus expectation, followed by projection to that subspace, is a bounded left inverse to $u \mapsto L(X(u))$. Compactness of L would make the identity on an infinite-dimensional Hilbert space compact, contradicting its orthonormal sequence. This is the elementary left-inverse mechanism behind the frozen general recovery obstruction.

At zero damping, unit cutoffs have divergent energy; under quadratic damping, the particular field has finite energy and a valid divergence. Nevertheless the underlying observation family still contains every $B_n(t)$ and retains Theorem 5.1. The two conclusions concern different operators and do not contradict one another.

6 Compact deterministic synthesis does not give a stochastic trace

The failed linear-rate example has a useful additional property. At fixed $t \geq 0$, define a *scalar* synthesis map on deterministic amplitudes,

$$T_t : \ell^2 \longrightarrow L^2(P), \qquad T_t c = \sum_{n\geq 0} c_n Y_n(t), \qquad \lambda_n = n + 1.$$

This is different from the Hilbert-valued random field $F_N(t)$: its output is a scalar Gaussian random variable, and its input is an arbitrary square-summable deterministic sequence. The original modal source isometry supplies T_t on exactly this coefficient space.

Proposition 6.1. *T_t is a compact bounded operator for every $t \geq 0$. For $t > 0$ it is not Hilbert–Schmidt and does not admit the unit-amplitude stochastic field of Theorem 1.1. If $T_t^{(N)} c = \sum_{n<N} c_n Y_n(t)$, then*

$$\|T_t - T_t^{(N)}\| \leq [2(N + 1)]^{-1/2}.$$

The sharper constant here is a direct consequence of the energy formula; the formal source records the sufficient bound $(N + 1)^{-1/2}$.

Proof. The source directions $h_{n,t}$ are orthogonal across n, so $\| \sum_{n\in A} c_n Y_n(t)\|_2^2 = \sum_{n\in A} c_n^2 V_{n+1}(t)$. Since $V_{n+1}(t) \leq 1/[2(n + 1)] \leq 1/2$, this defines a bounded operator on ℓ^2 by completion. For its tail,

$$\|(T_t - T_t^{(N)})c\|_2^2 = \sum_{n\geq N} c_n^2 V_{n+1}(t) \leq \frac{\|c\|_{\ell^2}^2}{2(N + 1)}.$$

Each $T_t^{(N)}$ has finite-dimensional range. Their norm convergence implies compactness: approximate the image of the unit ball to any positive tolerance by a bounded subset of a finite-dimensional range, which has a finite net; completeness turns total boundedness into relative

compactness. But for the standard coefficient basis ϵ_n,

$$\sum_n \|T_t\epsilon_n\|_2^2 = \sum_n V_{n+1}(t) \geq \frac{1-e^{-2t}}{2} \sum_n \frac{1}{n+1} = \infty \quad (t>0).$$

Thus T_t is not Hilbert–Schmidt. The same divergent sum is the cutoff field energy for $c_n = 1$, proving the last assertion. At $t = 0$, $T_0 = 0$ and both compactness and a zero trace are trivial. □

The constant sequence one is outside ℓ^2. Evaluating T_t there is not an authorized extension of its deterministic domain. Compactness controls a supremum of tail sizes; stochastic trace requires their sum. These are the classical compact-versus-Hilbert–Schmidt alternatives, now exhibited on the actual exponential modal construction.

Source correspondence: `SPDECompactModalSynthesis.synthesis_tail_bound`, `synthesis_compact`, `linearSynthesis_compact`, `linearSynthesis_infinite_trace`.

7 Wick centering and exact series energies

For $h \in \mathcal{H}$, define the variance-dependent Wick polynomial $\mathcal{W}_d(h)$ by

$$e^{uX(h)-u^2\|h\|^2/2} = \sum_{d\geq 0} \frac{u^d}{d!}\mathcal{W}_d(h). \tag{15}$$

For example $\mathcal{W}_0(h) = 1$, $\mathcal{W}_1(h) = X(h)$ and $\mathcal{W}_2(h) = X(h)^2 - \|h\|^2$. We use monic Hermite normalization, so the factorial in this definition is essential. Taking expectations of the product of two such exponentials gives $e^{uv\langle h,k\rangle}$. Comparing coefficients yields

$$\mathbb{E}\mathcal{W}_d(h) = 0 \ (d>0), \qquad \mathbb{E}[\mathcal{W}_d(h)\mathcal{W}_e(k)] = \mathbf{1}_{d=e}\, d!\langle h,k\rangle^d. \tag{16}$$

Coefficient comparison is justified by differentiating at zero: on compact parameter sets, polynomial derivatives times exponentials of a finite Gaussian vector are integrable. Thus this derivation concerns actual $L^2(P)$ random variables, not formal power series alone.

Theorem 7.1 (Exact Wick-series energy)**.** *Let (g_n) be pairwise orthogonal directions in $\mathcal{H}$, let $d \geq 1$ be an integer, and let $a_n \in \mathbb{R}$. Then*

$$S = \sum_n a_n \mathcal{W}_d(g_n) \quad \textit{converges in } L^2(P) \quad \Longleftrightarrow \quad \sum_n a_n^2 d!\|g_n\|^{2d} < \infty.$$

In that case the squared norm is exactly the latter sum, and the squared cutoff error is its corresponding tail. The limit is in homogeneous chaos d, belongs to the original closed gradient domain, and its canonical divergence recovery satisfies

$$\delta RS = S, \qquad \|RS\|_2^2 = \frac{1}{d}\|S\|_2^2.$$

Proof. Equation (16) makes distinct terms orthogonal. Pythagoras therefore gives the finite norm and every finite tail norm exactly. The nonnegative energy series converges precisely when those tails are Cauchy; completeness proves existence and necessity. The closed dth chaos contains the limit.

For completeness the gradient assertion uses its original graph. Differentiating (15) gives $D\mathcal{W}_d(g) = d\, g\mathcal{W}_{d-1}(g)$. For distinct orthogonal directions the resulting vector fields are orthogonal as well, and for a single term their squared norm is

$$d^2\|g\|^2(d-1)!\|g\|^{2(d-1)} = d\, d!\|g\|^{2d}.$$

Thus the finite partial sums and their gradients are Cauchy. Closedness places S in the original gradient domain with $\|DS\|_2^2 = d\|S\|_2^2$. On chaos d, the recovery already defined above is $R = D/d$, giving its exact energy. The finite product rule $\delta(gA) = X(g)A - \langle g, DA\rangle$, together with the recurrence for $d \geq 2$ $\mathcal{W}_d(g) = X(g)\mathcal{W}_{d-1}(g) - (d-1)\|g\|^2\mathcal{W}_{d-2}(g)$, gives $\delta DS_N = dS_N$ (the $d = 1$ case uses $\delta g = X(g)$); closedness of the original adjoint, applied to $DS_N/d \to DS/d$ and $S_N \to S$, proves $\delta RS = S$. □

Source correspondence: `SPDEWickModes.wickSeries_summable_iff`, `finite_wick_energy`, `wickResponse_energy`, `wickResponse_recovery_graph`, `wickResponse_recovery_energy`.

The explicit derivative-domain proof is a human reconstruction from the same finite Wick identities and original closed graph.

There are two concrete applications, whose normalizations should not be conflated. First take independent unit normals $G_n = B_n(1)$ and $a_n = 1/(n+1)$. The series

$$\sum_n \frac{G_n^2 - 1}{n+1} \tag{17}$$

converges in L^2, with variance $2\sum_n (n+1)^{-2}$. Its uncentered version does not: the means of its partial sums are the divergent harmonic sums, whereas L^2 convergence would imply convergence of expectations by Cauchy–Schwarz. The deterministic subtractions in (17) cannot be taken as a separately convergent series.

Source correspondence: `SPDEWickModes.actual_centered_squares_summable`, `actual_raw_squares_not_summable`.

Second use the exponentially damped directions $g_n = h_{n,t}$ at $\lambda_n = n+1$ and take $a_n = 1$. For every integer $d \geq 2$,

$$\sum_n \mathcal{W}_d(h_{n,t}) \quad \text{converges in } L^2, \qquad \|S_d(t)\|_2^2 = d!\sum_n V_{n+1}(t)^d < \infty,$$

since $V_{n+1}(t)^d \leq [2(n+1)]^{-d}$. At $d = 2$ the raw squares $\sum_n Y_n(t)^2$ again have divergent expectations for $t > 0$. This is a direct instantiation of Theorem 7.1, rather than a separately named positive-time theorem in the formal source. It constructs a scalar centered observable. It does not create the missing unit-amplitude Hilbert-valued first-chaos field or a nonlinear stochastic PDE solution. Its covariance energy is a power of the modal variance, which explains why its summability differs.

7.1 Arbitrary observations retain a full projected Gram matrix

Let (o_α) be any family in $\mathcal{H}$, let $K = \overline{\operatorname{span}}\{o_\alpha\}$, and let P_K be its orthogonal projection. Write $\mathcal{A}$ for the completed sigma algebra generated by the actual Gaussian variables $X(o_\alpha)$. The original pairwise orthogonality of (g_n) need not survive P_K. This changes the energy calculation, although it does not prevent convergence of a predicted response.

Lemma 7.2. *For every $g \in \mathcal{H}$ and integer $d \geq 0$,*

$$\mathbb{E}[\mathcal{W}_d(g) \mid \mathcal{A}] = \mathcal{W}_d(P_K g).$$

In particular the variance in the Wick polynomial on the right is $\|P_K g\|^2$, not $\|g\|^2$.

Proof. Put $v = P_K g$ and $r = g - v$. Approximation of v by finite linear combinations of observations, followed by L^2 convergence and an almost surely convergent subsequence, makes $X(v)$ measurable for $\mathcal{A}$. The variable $X(r)$ has zero covariance with every finite vector of observed coordinates. Joint Gaussianity gives independence from each such vector; the cylinder-event monotone class argument extends independence to their generated sigma algebra and its completion. Also $\langle v, r \rangle = 0$, so $\|g\|^2 = \|v\|^2 + \|r\|^2$. Hence for real z,

$$\mathbb{E}[e^{zX(g) - z^2\|g\|^2/2} \mid \mathcal{A}] = e^{zX(v) - z^2\|v\|^2/2}.$$

Both sides can be differentiated any fixed number of times at zero as L^2 variables. On a compact z interval each derivative is bounded by a fixed polynomial in the Gaussian coordinates times $e^{C(|X(v)|+|X(r)|)+C}$, whose square is integrable. Conditional expectation is bounded on L^2, so differentiation commutes with it. The generating function gives the claim. □

Theorem 7.3. *Assume the hypotheses and finite energy of Theorem 7.1. Set $v_n = P_K g_n$. Then*

$$\widehat{S} := \mathbb{E}[S \mid \mathcal{A}] = \sum_n a_n \mathcal{W}_d(v_n) \quad \text{in } L^2(P).$$

For every finite index set F, its partial-sum energy is exactly

$$G_F = d! \sum_{i,j \in F} a_i a_j \langle v_i, v_j \rangle^d, \qquad 0 \leq G_F \leq \sum_{i \in F} a_i^2 d! \|g_i\|^{2d}.$$

As finite sets increase to all indices, $G_F \to \|\widehat{S}\|_2^2$. The canonical original divergence recovery obeys

$$\delta R\widehat{S} = \widehat{S}, \qquad \|R\widehat{S}\|_2^2 = d^{-1}\|\widehat{S}\|_2^2.$$

Proof. Apply the bounded conditional expectation to the convergent finite partial sums of S and use Lemma 7.2 term by term. This proves convergence over arbitrary finite cutoffs, without an orthogonality assertion for (v_n). Expanding the finite squared norm and using (16) gives G_F. Its lower bound is a squared norm, and its upper bound is the contraction inequality for the conditional expectation of the original orthogonal finite sum. Norm continuity proves its limit. For any finite tail F the same argument gives

$$\Big\| \sum_{i \in F} a_i \mathcal{W}_d(v_i) \Big\|_2^2 \leq \sum_{i \in F} a_i^2 d! \|g_i\|^{2d},$$

so no absolute convergence of an infinite double sum is needed. Every predicted finite sum lies in homogeneous chaos d, and that chaos is closed. The inverse-number recovery energy is therefore $d^{-1}\|\widehat{S}\|_2^2$. Its original adjoint graph follows either from the previously proved recovery on centered L^2 or by bounded recovery convergence and closedness of δ on the finite-sum graph. The target is centered because $d \geq 1$. □

Source correspondence: `GaussianWickConditionalProjection.conditional_wick_projection`, `conditional_wick_energy`.
`SPDEWickPrediction.finite_prediction_gram`, `projected_gram_bounds`, `predicted_response_hasSum`, `projected_gram_limit`, `predicted_response_recovery_graph`.

A two-mode calculation shows why the cross terms matter. Take orthonormal u, v, observe only $X(u+v)$, and put $w = (u+v)/2$. Projection onto that line gives $P_K u = P_K v = w$ and $\|w\|^2 = 1/2$. Consequently

$$\mathbb{E}[\mathcal{W}_d(u) - \mathcal{W}_d(v) \mid \mathcal{A}] = 0, \quad \big\|\mathbb{E}[\mathcal{W}_d(u) + \mathcal{W}_d(v) \mid \mathcal{A}]\big\|_2^2 = 4d!2^{-d}.$$

Indeed the two forecasts are the same variable $\mathcal{W}_d(w)$, with squared norm $d!2^{-d}$. Keeping only diagonal terms would assign the difference the strictly positive energy $2d!2^{-d}$ and assign the sum only half its actual energy. Each original sum or difference has energy $2d!$, so the sum's retained fraction is 2^{1-d}. With actual independent Brownian unit-time modes and $d = 2$, its predicted energy is exactly two. These are the mapped two-mode and actual Brownian calibration statements, not an assertion that arbitrary projection preserves independence.

Gaussian Wick prediction and its tensor interpretation are classical. Hu–Yan [2, Section 2, (2.6)–(2.7)] give the monic-Hermite multiple-integral normalization and isometry used by the Gram formula. The conditional formula itself has been proved above, rather than imported from a theorem about a scalar second-quantization parameter. None of these scalar observable predictions constructs a nonlinear stochastic PDE solution or restores a missing first-chaos stochastic trace.

8 Initial scaling and positive-time roughness

A final exact example separates chaos degree from temporal roughness. For any square-summable $a = (a_n)$ and integer $d \geq 1$, set

$$Z_d(t) = \sum_n a_n \mathcal{W}_d(\iota_n \mathbf{1}_{(0,t]}), \qquad C_d = d! \sum_n a_n^2.$$

This definition uses undamped Brownian directions and is separate from the exponentially damped $S_d(t)$. Theorem 7.1 gives $\|Z_d(t)\|_2^2 = C_d t^d$ and $Z_d(0) = 0$.

For $0 \leq s \leq t$, independent Brownian increments give

$$\mathbb{E}[Z_d(t) \mid \mathcal{A}_s] = Z_d(s). \tag{18}$$

Here $\mathcal{A}_s$ is the full modal past defined earlier, not only the present values. To verify this point explicitly, the Gaussian increment $B_n(t) - B_n(s)$ has zero covariance with every $B_m(r)$, $r \leq s$. Joint Gaussianity gives independence from each finite past vector, and a monotone-class argument gives independence from their entire generated sigma-algebra. Factoring the exponential in (15) into its past and increment factors shows that the conditional expectation of its increment factor is one. Differentiation at zero, justified by Gaussian exponential domination, proves the finite Wick identity. Conditional expectation is an L^2 contraction, so passing the convergent series proves (18).

Orthogonal projection onto the observed past now gives

$$\mathbb{E}[Z_d(t) Z_d(s)] = C_d s^d, \qquad \|Z_d(t) - Z_d(s)\|_2^2 = C_d(t^d - s^d), \quad s \leq t. \tag{19}$$

Indeed $Z_d(s)$ is observed at s, so its pairing with $Z_d(t)$ equals its pairing with the conditional expectation in (18). Expansion of the squared norm proves the second identity. It also proves mean-square continuity on all $[0, \infty)$ without a sample-path assertion.

Source correspondence: `SPDEWickBrownianPrediction.full_past_prediction`, `SPDEWickTemporalGeometry.response_prediction_between`, `response_energy_scale`, `response_increment_energy`.

At the origin the norm is $\sqrt{C_d}\,t^{d/2}$. At any fixed $s > 0$, however, if $C_d > 0$,

$$\frac{\|Z_d(s+h) - Z_d(s)\|_2^2}{h} = C_d \frac{(s+h)^d - s^d}{h} \longrightarrow C_d d s^{d-1} > 0.$$

Thus a high initial power does not give the corresponding higher increment power at positive times. In the formal harmonic quadratic example $a_n = (n+1)^{-1}$, $C_2 \geq 2$. At $s = 1/2$ the exact squared increment is $C_2(h + h^2) \geq 2h$. For every $\alpha > 1$, every real K and every $\eta > 0$, sufficiently small $0 < h < \min(\eta, 1/2)$ therefore violate a proposed bound $\|Z_2(1/2+h) - Z_2(1/2)\|_2^2 \leq Kh^\alpha$. For $K \leq 0$ the violation is immediate; for $K > 0$ use $Kh^{\alpha-1} \to 0$. This is a statement about exact L^2 increments, not a claimed optimal almost-sure Hölder exponent.

Source correspondence: `SPDEWickTemporalGeometry.harmonic_positive_time_increment_lower`, `no_higher_positive_time_power`.

Wick centering, chaos orthogonality and Brownian Hermite martingales are classical. Baudoin's primary lecture [1] derives the exponential/Hermite representation and the chaos derivative-energy identity used here. Its Hermite normalization includes $1/d!$ in the polynomial itself; ours is the monic normalization in (15). Neither the centered-series examples nor their initial-versus-positive-time scaling are claimed as new principles. They give explicit, source-traceable tests against transferring a compactness, first-chaos trace or initial-scaling statement to a different stochastic object.

9 Covariance powers do not identify the process law

The preceding Wick martingales give an actual realization of integer entrywise covariance powers. If $X(h_t)$ is any Gaussian source family, define

$$V_q(t) = \frac{\mathcal{W}_q(h_t)}{\sqrt{q!}}, \qquad q \geq 1.$$

The proved Wick pairing identity gives

$$\mathbb{E}[V_q(s)V_q(t)] = \langle h_s, h_t\rangle^q.$$

For any finite times and real coefficients c_i it follows that

$$\sum_{i,j} c_i c_j \langle h_{t_i}, h_{t_j}\rangle^q = \left\| \sum_i c_i V_q(t_i) \right\|_2^2 \geq 0.$$

Thus the entrywise power is a genuine positive semidefinite Gram matrix. For fractional Brownian directions the exact same identity is the content of `FractionalWickCovariance`; it is a statement about actual finite covariances and original Gaussian derivatives, not a new fractional-time generator or a Hermite-process construction.

For Brownian $h_t = \mathbf{1}_{(0,t]}$, take independent actual modal Brownian motions B_0, B_1 from the initial construction and set

$$V_q(t) = \frac{\mathcal{W}_q(\iota_0 \mathbf{1}_{(0,t]})}{\sqrt{q!}}, \qquad G_q(t) = B_1(t^q), \qquad t \geq 0.$$

Their paths are continuous, both are centered martingales for their respective Brownian pasts, and both have covariance $\min(s,t)^q$. For V_q the martingale proof is the full-past Wick identity above; for G_q it is Brownian independence on the deterministic increasing clock t^q. No equality of their laws follows.

Indeed, already at $q = 2$,

$$V_2(t) = \frac{B_0(t)^2 - t}{\sqrt{2}}, \qquad \mathbb{E}V_2(t)^2 = t^2, \qquad \mathbb{E}V_2(t)^4 = 15t^4,$$

whereas $\mathbb{E}G_2(t)^4 = 3t^4$. To verify the former fourth moment, expand $(B_0(t)^2 - t)^4$ and use the Gaussian even moments $1, 3, 15, 105$ in degrees $2, 4, 6, 8$; the numerator is $60t^4$, and division by $(\sqrt{2})^4 = 4$ gives $15t^4$. For $s > 0$, $h > 0$, let $\Delta = B_0(s+h) - B_0(s)$. The exact increment identity

$$V_2(s+h) - V_2(s) = \sqrt{2}B_0(s)\Delta + \frac{\Delta^2 - h}{\sqrt{2}}$$

and independent centered normal Δ give

$$\mathbb{E}[(V_2(s+h) - V_2(s))^2 \mid \mathcal{F}_s^{B_0}] = 2hB_0(s)^2 + h^2.$$

The cross term vanishes by the third normal moment. Since $B_0(s)^2 = \sqrt{2}V_2(s) + s$ is observed in the V_2 past, the tower property gives this same nonconstant conditional second moment on that smaller past. Thus its increments are not independent. The corresponding Gaussian increment of G_2 is independent of its past and has deterministic variance $(s+h)^2 - s^2 = 2sh + h^2$. Identical covariance and unconditional increment energies have therefore concealed different conditional laws. These explicit fourth-moment and conditional-variance comparisons are classical human deductions, not additional named formal theorems.

9.1 Exact finite innovations and generalized eigenvalues

At the integer times $1, \ldots, n$, the common covariance matrix is

$$C_k(i,j) = \min(i+1, j+1)^k, \qquad 0 \le i, j < n, \quad k \ge 1.$$

Write $L_{ij} = \mathbf{1}_{\{j \le i\}}$ and $d_k(j) = (j+1)^k - j^k > 0$. Telescoping the powers gives the exact factorization

$$C_k = L \, \mathrm{diag}(d_k(j)) L^\top.$$

It is a covariance factorization: the Gaussian process G_k has independent innovations with these variances. The non-Gaussian process with the same covariance need not have independent innovations, as the example just proved shows. In particular a finite covariance calculation is not a probabilistic independence argument.

The triangular matrix L is invertible, its inverse taking adjacent differences. Thus, for $k, l \ge 1$, the exact generalized eigenvalues of (C_k, C_l) are

$$\rho_j = \frac{d_k(j)}{d_l(j)}.$$

More explicitly, let $v_0 = e_0$ and $v_j = e_j - e_{j-1}$ for $j > 0$. Then $L^\top v_j = e_j$, so $C_k v_j = d_k(j) L e_j = \rho_j C_l v_j$. These n nonzero vectors form a basis. Congruence by L^{-1} also proves the sharp quadratic-form comparisons

$$\left(\min_j \rho_j\right) C_l \preceq C_k \preceq \left(\max_j \rho_j\right) C_l.$$

Here $A \preceq B$ means that $B - A$ has a nonnegative quadratic form. Indeed the diagonal congruence of $C_k - cC_l$ has entries $d_k(j) - cd_l(j)$; these are all nonnegative exactly when $c \le \min_j \rho_j$. The upper comparison is identical with the roles reversed, and the vectors v_j attain the endpoints. These sharp comparison statements are direct classical deductions from the verified factorization and generalized eigenpairs.

For $k = 3$, $l = 2$, $j = 1$, the comparison value is $7/3$, not the single innovation variance 7. Nor is C_k the ordinary matrix power of C_1. At $n = 2$,

$$C_1 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad C_2 = \begin{pmatrix} 1 & 1 \\ 1 & 4 \end{pmatrix}, \quad C_1^2 = \begin{pmatrix} 2 & 3 \\ 3 & 5 \end{pmatrix}.$$

Entrywise covariance powers, ordinary matrix powers, independent Gaussian innovation variances and generalized comparison ratios are four different quantities. None of these finite identities identifies a physical-time generator, curvature operator or scale-independent estimate. The underlying Wick and Gaussian mechanisms are classical, with [1] providing the previously credited chaos context.

10 Wick metric transfer and its antipodal boundary

Covariance powers also give an exact metric calculation. Retain the monic Wick convention of (15) and put $V_q(h) = \mathcal{W}_q(h)/\sqrt{q!}$, for an integer $q \ge 1$. The mixed Wick moment identity already proved gives

$$\|V_q(h) - V_q(k)\|_2^2 = \|h\|^{2q} + \|k\|^{2q} - 2\langle h, k\rangle^q. \tag{20}$$

This identity follows by expanding the squared norm and applying the same pairing formula to each of the three terms; it holds for all source directions, with no equal-variance assumption.

Proposition 10.1. *If $\|h\| = \|k\| = 1$ and $0 \le \rho := \langle h, k\rangle \le 1$, then*

$$\|h - k\|^2 \le \|V_q(h) - V_q(k)\|_2^2 \le q\|h - k\|^2.$$

At degree two and equal squared norm v, the exact formula is

$$\|V_2(h) - V_2(k)\|_2^2 = (v + \langle h, k\rangle)\|h - k\|^2.$$

In particular $V_2(h) = V_2(-h)$ as L^2 classes, so the global map is not injective when the source space is nonzero.

Proof. For unit directions, (20) is $2(1 - \rho^q)$, whereas $\|h - k\|^2 = 2(1 - \rho)$. Since

$$1 - \rho^q = (1 - \rho)\sum_{j=0}^{q-1} \rho^j, \qquad 1 \le \sum_{j=0}^{q-1} \rho^j \le q,$$

the bounds follow, including $\rho = 1$. For equal squared norm v, the degree-two formula is $2(v^2 - \langle h, k\rangle^2) = (v + \langle h, k\rangle)(2v - 2\langle h, k\rangle)$. Taking $k = -h$ makes this norm zero. If $h \ne 0$, its source distance from $-h$ is $2\|h\| > 0$, proving noninjectivity. More generally, (20) gives this antipodal identification at every even degree. Consequently no positive global lower metric constant on the full unit sphere follows from the positive-correlation bounds. □

Here is a precise fractional application, with a labelled additional human asymptotic deduction. For $0 < H < 1$, let h_t be the actual fractional Gaussian directions with

$$\langle h_s, h_t \rangle = \tfrac{1}{2}(s^{2H} + t^{2H} - |t-s|^{2H}), \qquad \|h_t\| = t^H.$$

For $t > 0$ define $u_t = t^{-H} h_t$. These are unit directions. Their inner products are nonnegative: for $0 \le s \le t$, $(t-s)^{2H} \le t^{2H}$, so the covariance is nonnegative. Cauchy–Schwarz gives the upper bound one. The proposition therefore applies to the normalized fractional coordinates $V_q(u_t)$. It does not apply at zero by assigning a value to $0^{-H} h_0$.

For clarity the exact positive-time roughness can be read from these formulas, without a regularity theorem for sample paths. Fix $0 < a < b < \infty$. For $a \le s < t \le b$ and $d = t - s$,

$$\|u_t - u_s\|^2 = \frac{d^{2H} - (t^H - s^H)^2}{s^H t^H}. \tag{21}$$

Indeed expansion of the normalized covariance gives this equality. The mean-value theorem bounds $|t^H - s^H| \le Ld$ with $L = Ha^{H-1}$. Choose $d_0 > 0$ so that $L^2 d_0^{2-2H} \le 1/2$, possible since $H < 1$. For $0 < d \le d_0$, (21) and the proposition give the full two-sided bounds

$$\frac{d^{2H}}{2b^{2H}} \le \|u_t - u_s\|^2 \le \|V_q(u_t) - V_q(u_s)\|_2^2 \le q\|u_t - u_s\|^2 \le \frac{q\, d^{2H}}{a^{2H}}.$$

In particular every fixed degree has the same local mean-square increment exponent $2H$ on positive-time compact intervals. An upper bound of order d^β with $\beta > 2H$ cannot hold uniformly there: fix $s \in (a, b)$, take $t = s + d$ and divide the positive lower bound by d^β. It diverges as $d \downarrow 0$.

The source maps the normalized fractional metric comparison; the last compact-interval exponent calculation is the human deduction just proved. Normalization changes the process and excludes time zero. This is a statement about L^2 increments, not an optimal sample-path Hölder theorem. Pointwise Wick polynomials of one fractional Gaussian family are not identified with a separately constructed Hermite process, nor with a positive Markov generator. The calculation uses the classical Wick pairing mechanism credited above, without a new covariance or roughness principle claim.

11 Formal correspondence and prior theory

The source authority is frozen verified commit `bab688bb6140cd6316c80680095a914ea0d346db`. The countable isonormal modes and actual orthogonal series are in `SPDECountableModes`; exponential profiles and reciprocal-rate criteria are in `SPDEExponentialForcing`. The actual Hilbert-valued field and divergence application are in `SPDEHilbertExponentialForcing`; time continuity is `SPDEExponentialTimeContinuity`. The completed original divergence, second-chaos and projection-energy results are `GaussianLinearFieldDivergenceCompletion`. The full observed mode obstruction is `SPDEObservationRecoveryCompactness`.
The theorem map distinguishes exact declarations from elementary human reorganizations: Hilbert–Schmidt tensors present the formal closed linear-field space, the finite fourth-moment proof reconstructs its quadratic completion, and cutoff errors follow from the exact orthogonal energy identities. The maximal Gaussian adjoint and coefficient-space geometry remain those of the frozen construction. No coefficient alignment is silently added to the application. The new compact synthesis, Wick-series energy and Brownian prediction/temporal identities are linked to

their original source declarations as well. The accompanying `verify_wick_modes.py` passes 26 exact rational polynomial checks, including conditional Hermite identities through degree eight, correlated Wick covariances through degree six and finite linear-rate tail bounds. Infinite convergence, graph closure and asymptotic exclusions rely on the printed proofs.
Hairer's primary notes establish the classical Hilbert–Schmidt stochastic-integral and stochastic-convolution setting. The scalar reciprocal-rate threshold, finite-chaos divergence and infinite-mode compactness obstruction are presented here as a consolidated formalization companion. Gaussian derivative/adjoint conventions are also compared with the primary treatment [4]. No new stochastic PDE existence theorem, optimal sample-path regularity, or priority claim is made. A samplewise space-time realization, nonlinear equation or stronger convergence topology requires its own hypotheses and proof. Full baseline build and axiom evidence was hash-checked rather than repeated at the expense of active research resources.

References

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[4] T. Sottinen and L. Viitasaari, *Stochastic analysis of Gaussian processes via Fredholm representation*, Int. J. Stoch. Anal. (2016), article 8694365. doi:10.1155/2016/8694365.